

Hands-On Labs - Generative AI for Image and Video Creation

WSQ Course Code: TGS-2020505925 | Version v11.0 | 6 September 2026 Conducted by: Tertiary Infotech Academy Pte Ltd (UEN 201200696W)

Twelve labs across the five delivery topics. Every folder is independently usable: it carries its own steps, its own mock data, its own local media and its own prompt pack, and needs no network access once the Python dependencies are installed.

Lab	Title	Topic	Outcome	A	K	In-class
01	Vision-System Needs Analysis and Generative Pipeline Specification	01	ELO1	A1	K1, K2	within LU1 facilitation
02	Image and Video Data Representation	02	ELO2	A2	K3	15 min
03	Prompt Experiment Matrix for Controlled Image Generation	02	ELO2	A2	K3, K4	15 min
04	Masking, Inpainting, Outpainting and Background Swap	02	ELO2	A2	K3, K4	15 min
05	Filtering, Enhancement and Restoration Measured with PSNR and SSIM	02	ELO2	A2	K4	15 min
06	Local Features: Colour Segmentation, Canny Edges and Corner Keypoints	03	ELO3	A4	K5, K6	30 min
07	Global Descriptors, Template Matching and Brand-Consistency Scoring	03	ELO3	A3, A4	K7	30 min
08	Object Detection and Quantitative Evaluation	04	ELO4	A5	K8, K9	60 min

Lab	Title	Topic	Outcome	A	K	In-class
09	Storyboard to Video Prompt Pack and Deterministic Animatic	04	ELO5	A6	K10	30 min
10	Motion, Tracking and Temporal Continuity Measurement	04	ELO5	A6	K10	60 min
11	Captions, Overlays, Safe Areas and Export Profiles	04	ELO5	A6	K10	30 min
12	Cloud-Edge Architecture Design and Measured Trade-offs	05	ELO6	A7, A8	K11, K12, K13	60 min

Total scheduled practical time: 360 minutes (6 hours) - matching the 360-minute practical allocation in the approved course proposal.

Coverage

Code	Statement	Labs
A1	Identify the needs of vision systems technology in industrial applications	01
A2	Apply the principles of processing, filtering and analysis methods for video data	02, 03, 04, 05
A3	Analyse global feature descriptions	07
A4	Design and implement feature extraction and representation methods	06, 07
A5	Design and apply machine-learning based methods for object detection, object tracking and activity recognition	08
A6	Design and apply video analytics algorithms for high-level video analytics tasks	09, 10, 11
A7	Design the architecture of applied vision systems	12
A8	Design, develop and evaluate edge-based and cloud-based systems	12
K1	Vision system concepts	01
K2	Business applications of vision systems	01
K3	Methods to represent image and video data	02, 03, 04

Code	Statement	Labs
K4	Image and video processing, filtering and transformation methods	03, 04, 05
K5	Feature extraction and representation techniques	06
K6	Local feature descriptions, edge, colour, texture and motion	06
K7	Global feature descriptions, statistical and geometrical methods	07
K8	Deep learning concepts	08
K9	Object segmentation, detection and recognition	08
K10	Activity tracking, generative models, scene understanding and event discovery	09, 10, 11
K11	Vision system architecture	12
K12	Vision communication protocols	12
K13	Real-world design constraints and solution options	12

Environment

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install "opencv-python>=4.10" numpy
```

No lab requires a paid service. Where a generative model can optionally be used, the lab supplies a local reference set so the deliverable can be completed offline, and asks you to record which path you took.

Provenance rules that apply to every lab

1. A SIMULATED asset is a deterministic OpenCV rendering. It is never described as a photograph and never as generative-model output.
2. An AI-generated reference is copied unchanged and travels with a PROVENANCE.md recording the tool, date and prompt.
3. A real prerecorded sequence records its source URL, retrieval date and SHA-256.
4. A deterministic rendered animatic is labelled ANIMATIC on every frame and is never submitted as a generated film.
5. Anything you generate yourself is recorded with the model ID, the prompt version and the watermark expectation.

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